

K-Nearest Neighbour in ML

1) What is KNN?
Ans: KNN is a supervised machine learning used for both classification and regression. KNN predicts ~~for~~ on the basis of majority for classification and on the basis of Average for regression.

KNN works based on ~~the~~ & similar data points are likely to have similar outcomes.

2) Important Terms!

i) k = number of neighbours we consider.

Example:

$k=3 \rightarrow$ look at 3 nearest points

$k=5 \rightarrow$ look at 5 " "

ii) Distance $\rightarrow$ measure how close two data points are

Smaller distance = closer neighbour

Neighbour $\rightarrow$ A nearby data points.

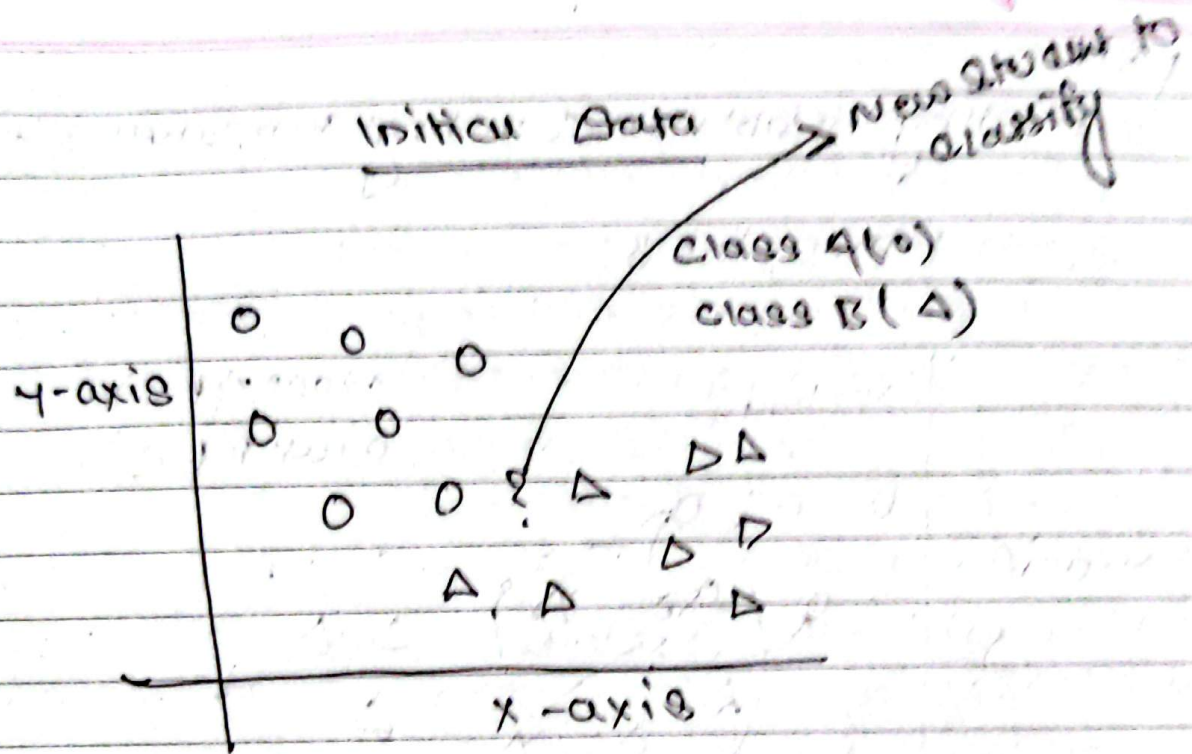
3) How KNN works!

Suppose we want to classify a new student as pass or fail.

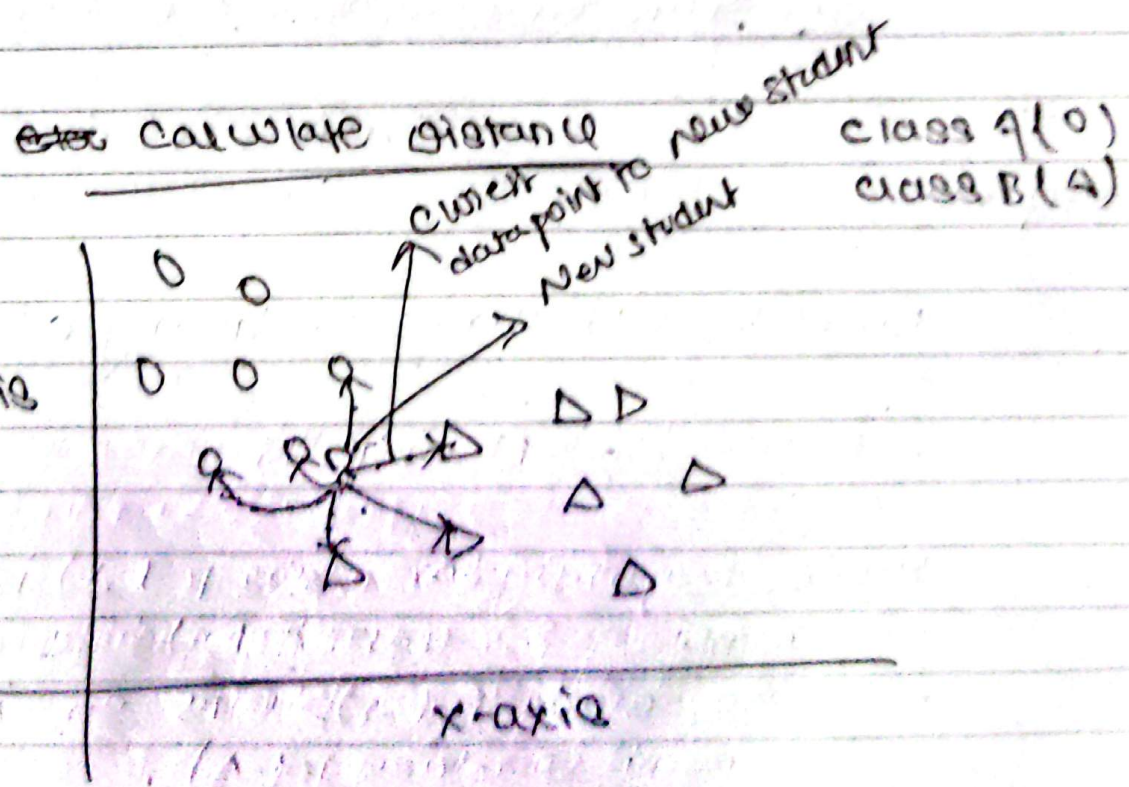
New student $\rightarrow$ study hour $\rightarrow 5.5$
 $k=3$

Step 1: calculate distance from new student to all ~~closest~~ existing students.

Step 2: select the 3 closest students.



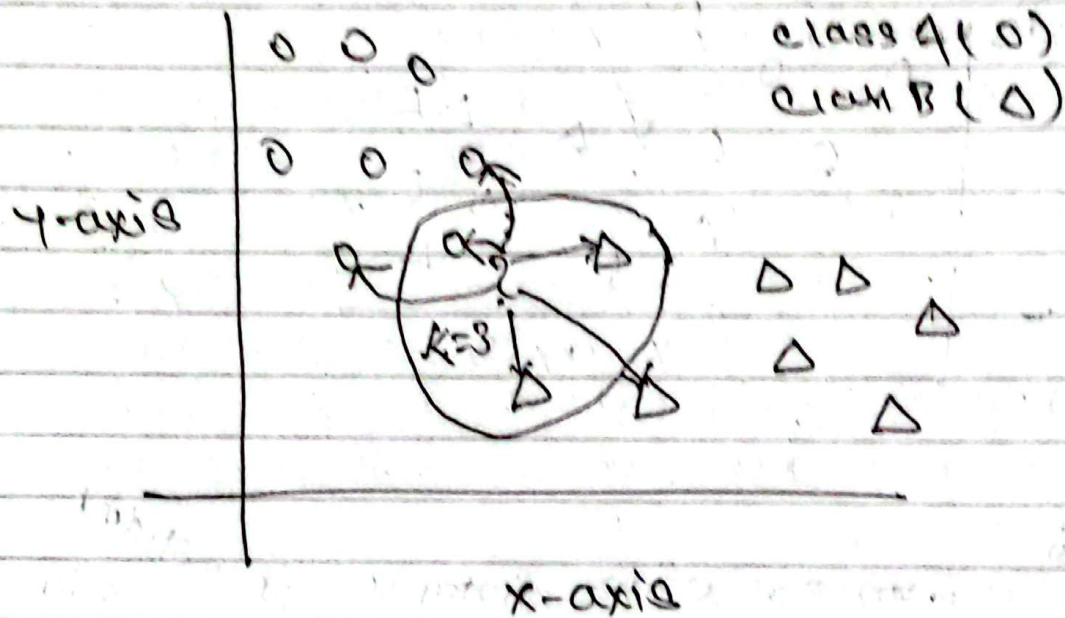
Step 4:



Step 5%

Finding Neighbour & voting For labels

Since, $k = 3$



Step 6%

Is new point circle or triangle?

(?) New point

Since, majority for class B (Δ) is 2
majority for class A (O) is 1
So, class B (Δ) wins so new
point is class B (Δ).

4. KNN classification

For classification, KNN uses majority voting

Eg. $k=5$

Neighbour 1 = Disease

" 2 = Disease

" 3 = No disease

" 4 = Disease

" 5 = No disease

Count

Disease = 3

No disease = 2

prediction $\Rightarrow$ Disease

5. KNN regression

For regression KNN uses the average value of nearest neighbour.

Eg:

$k=3$

House price,

50 lakh

55 "

60 lakh

prediction

$$(50 + 55 + 60) / 3 = 55 \text{ lakh}$$

prediction = 55 lakh

6) Why Feature scaling is important in KNN?

KNN uses distance, so features should be ~~also~~ usually be on a similar scale.

For eg!

Age = 20 - 80

Income = 20000 - 200000

Income has much larger value and can dominate distance $\therefore$ we use,

Standard scalar or minMax scalar

7) choosing k

Small k $\rightarrow$ sensitive to noise $\rightarrow$ overfitting
 large k $\rightarrow$ too general $\rightarrow$ underfitting

We generally we analyze different k values and choose the best one.